

CELIA RUBIO-MADRIGAL

 celrm.github.io

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EXPERIENCE



- **Doctoral researcher** – CISPA Helmholtz Center for Information Security **2023-now**
 - Relational Machine Learning Lab  – Supervisor: Dr. Rebekka Burkholz 
- **RA: Departmental collaboration** – Ministerio de Educación & Universidad Complutense de Madrid **2021-22**
 - Identification of Boolean functions that need big circuits  – Supervisor: Dr. Ismael Rodríguez Laguna 
- **REU: Introduction to research ICMAT-Severo Ochoa** – Instituto de Ciencias Matemáticas (ICMAT) **2021**
 - Differential geometry methods in ML applications  – Supervisor: Dr. David Martín de Diego 

EDUCATION



- **PhD in Computer Science** – Universität des Saarlandes (Saarbrücken, Germany) **2023-now**
- **MSc in Advanced CS with AI** – University of Strathclyde (Glasgow, UK) – 8.57/10 (Distinction) **2022-23**
 - Fully funded by "la Caixa" Foundation Fellowship  – Earned the Department's Best Overall Prize.
- **BSc in Mathematics** – Universidad Complutense de Madrid (Spain) – 9.08/10 **2017-22**
- **BSc in Computer Science** – Universidad Complutense de Madrid (Spain) – 9.16/10 **2017-22**

PUBLICATIONS



- **Peer-reviewed publications** * Equal contribution.
 - Fixed Aggregation Features can rival GNNs. C.R.M., R. Burkholz. 20th Women in Machine Learning Workshop (WiML) at 39th NeurIPS, 2025.
 - GNNs getting ComFy: Community and feature similarity guided rewiring. C.R.M.*, A. Jamadandi*, R. Burkholz. 13th International Conference on Learning Representations (ICLR), 2025. 
 - Spectral graph pruning against over-squashing and over-smoothing. A. Jamadandi*, C.R.M.*, R. Burkholz. 38th Conference on Neural Information Processing Systems (NeurIPS), 2024. 
- **Under review**
 - Boosting for predictive sufficiency. A.G. Reddy, R. Verma, C.R.M., R. Burkholz, K. Muandet. 2025.
 - When shift happens-Confounding is to blame. A.G. Reddy, C.R.M., R. Burkholz, K. Muandet. arXiv, 2025. 
 - HAM: A hyperbolic step to regulate implicit bias. T. Jacobs, A. Gadhikar, C.R.M., R. Burkholz. arXiv, 2025. 

LANGUAGES



Spanish – Native

English – TOEFL iBT 110/120 (2021)

German – Goethe-Zertifikat B1 (2017)

TALKS



- **Invited talks** – *Seminars and reading groups*
 - 🎧 Invited talk: Graph Learning on Wednesdays (GLOW) community: *online* 📺 Oct 2025
 - 🎧 Visit & invited talk: Seminar of the TUWien ML Research Unit: *Vienna, Austria* May 2025
 - 🎧 Invited talk: Cohere Labs / Cohere For AI - Open Science Community: *online* 📺 Feb 2025
- **Contributed talks** – *Conferences and workshops*
 - 🎧 Lightning talk: Workshop on Geometry, Topology, and ML (GTML): *Leipzig, Germany* Nov 2025
 - 🎧 Two lightning talks: Workshop on Mining and Learning with Graphs (MLG): *Porto, Portugal* Sep 2025
 - 🎧 Lightning talk: International Conference on Network Science (NetSci): *Maastricht, Netherlands* Jun 2025
 - 🎧 Contributed talk: Helmholtz AI Conference: AI for Science (HAICON): *Düsseldorf, Germany* Jun 2024

AWARDS



- **Postgraduate**
 - Best Overall Performance Prize: *University of Strathclyde's top student in MSc Advanced CS* 2023
 - 🎧 Nova III Student List: *Spain's top 10 talents under 25 in Computer Science* 2023
 - 9th Heidelberg Laureate Forum, with interviews: 📺 HLFF Spotlight, 📺 Tribuna Complutense 2022
 - 🎧 "la Caixa" Foundation Fellowship, with interviews: 📺 El País, 📺 El Plural, 📺 TeleMadrid 2022
- **Undergraduate**
 - 🎧 Aula de Verano "Blas Cabrera" for Spain's top undergrads: *Intl. Univ. Menéndez Pelayo (UIMP)* 2022
 - 🎧 SWERC contestant (competitive programming): *ICPC Southwestern Europe Regional Contest in Italy* 2022
 - Becas de Excelencia: *Excellence Grants for undergrads with >9/10 in Madrid (2100€)* 2018-22
 - 🎧 EGMO contestant representing Spain: *European Girls' Mathematical Olympiad in Switzerland* 2017

TEACHING



- **Master's and Bachelor's thesis advising** – *CISPA & Universität des Saarlandes* 2024-now
- **Teaching Assistance: Laboratory demonstrator (7h/week)** – *University of Strathclyde* 2023
- "CS310 Foundations of AI" for Year 3 BSc Computer Science – Lecturer: Dr. Andrew Abel 📧

SERVICE



- **Academic service** – *Reviewer for MLG @ ECML PKDD 2025, ICLR 2026, CPAL 2026.* 2024-now
- **Mathematical Olympiad service**
 - 🎧 IT at European Girls' Mathematical Olympiad (EGMO) with Joseph Samuel Myers 📧 2023-now
 - Deputy Leader of Spain at EGMO 2022 and PAGMO 2021. 2021-22
- **Student representation** – *Facultad de Informática, Universidad Complutense de Madrid* 2020-22
 - Secretary of the Fdl Student Council 2021-22. – Member of the Fdl Faculty Board 2020-22.
- **LaTeX package development: ddphonism** 2019
 - 🎧 Generates 12-tone music diagrams with TikZ; included in MiKTeX and TeXLive.